

Highlights

INTERVIEW: Towards a Unified VR-Capable Interview Environment

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- A unified interview environment supports both VR headset and web-based platforms.
- Questionnaires, avatars and environments are customizable across platforms.
- Multimodal signals are stored as linked nodes for relational post-processing.
- A VR study with 27 participants shows a usable and stable interview setting.
- Most participants show behavioral shifts without self-reporting discomfort.

INTERVIEW: Towards a Unified VR-Capable Interview Environment

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ABSTRACT

Since surveys continue to face challenges such as attrition, recruitment difficulties, and rising costs, researchers are exploring various technological alternatives. One such alternative concerns the shift toward digital interview settings, such as video interviews. A second development builds on the growing capabilities of XR and, especially, VR systems, in which one or more interview partners are represented as virtual avatars. A third development concerns hypertext systems as an approach to modeling and representing extensible textual data structures based on the networked organization of information. Leveraging these three shifts (multimodality, XR, and informational networking), we introduce INTERVIEW: a unified interview environment that takes advantage of the multimodality of XR. It enables users to participate in interviews from virtually any platform and ensures that a wide range of multimodal signals exchanged in digital interview settings are displayed, stored in a database, networked, and made available for post-processing. INTERVIEW provides an environment specifically designed for interview-driven survey research and enables scientific insights to be derived from the analysis of structured multimodal data. We evaluate INTERVIEW through a study involving 27 participants who took part in interviews while wearing a VR headset. Our evaluation shows that INTERVIEW is usable and stable. The multimodal signals it captures - gaze, speech and response timing - reveal a measurable behavioral footprint of discomfort, in response to sensitive questionnaire items, even among participants who did not report feeling uncomfortable. This indicates that self-reporting alone would substantially underestimate such effects.

1. Introduction

Contemporary survey research faces increasing challenges related to participant recruitment (Savard and Kilpatrick, 2022), panel mortality (Lynn, 2018; Kunz, Börlin and Gummer, 2025) and rising costs (Minderop, 2023). As digital technologies have become ubiquitous among the general population, new opportunities for conducting interviews have emerged (Archibald, Ambagtsheer, Casey and Lawless, 2019), a shift further accelerated by the COVID-19 pandemic and the normalization of remote forms of interaction (Agostino, Arnaboldi and Lema, 2021; Waizenegger, McKenna, Cai and Bendz, 2020). To address the systemic issues that (panel) surveys have faced in recent years, various options are being explored, including recruitment strategies (Höhne, Claassen, Kühne and Zindel, 2026; Milbourn, Campbell, Clark, Darrah, Flaig, Kirby, McCartney, Mitchell, Robertson, Richmond et al., 2025), incentive schemes and interview modes (Sakshaug, Beste and Trappmann, 2023; Coffey, Maslovskaya and McPhee, 2024), and automation (Haensch, Weiß, Steins, Chyrva and Bitz, 2022; Gweon and Schonlau, 2024), with the broader aim of reducing operational costs (Coffey et al., 2024). Beyond replicating the traditional face-to-face interview process, these technologies introduce new modalities, such as spatial context, embodiment, and multimodal communication, that can be both designed for and captured. Instead of treating these aspects as a by-product, we consider them a central opportunity in a new interview environment, allowing us to further contextualize the respondents' answers by capturing additional behavioral signals.

While Computer-Assisted Live Video Interview (CALVI) frameworks have been shown to support simple question-answer exchanges (Durrant, Kocar, Brown, Hanson, Sanchez, Wood, Taylor, Tsantani and Huskinson, 2024; O'Reilly, Hubbard, Lessler, Biemer and Turner, 1994), they still model the interview as a linear dialogue. Hypertext theory,

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in contrast, emphasizes representing information as interconnected structures rather than linear sequences (Conklin, 1987; Halasz, Schwartz, Grønbaek and Trigg, 1994). If we extend this concept to real-time questionnaires and surveys, interview data shifts from isolated question-answer pairs toward a network of contextual and embodied aspects of interaction. Importantly, such a hypertextual interview structure interlinks multimodal artifacts, such as spatial configuration, avatar representation, and interaction dynamics, by conceptualizing them as nodes and relations within a network rather than as external metadata. During an interview, these relations can be captured alongside the interaction itself without altering its flow. This approach can be supported by multimodal eXtended Reality (XR) environments as well as by video interview environments, both of which differ in terms of accessibility and data richness. We identify two key dimensions.

The first dimension addresses interview design and system capabilities. Compared to a CALVI setup, XR-based interview environments allow for a broader spatial configuration, which is important because the interview space is widely acknowledged to shape the interview process (Elwood and Martin, 2000). These include customizable avatars and environments (Rzeszewski, Osborne, Jones, Evans and Weitkamp, 2024), which are not exclusive to XR, as well as the dynamic representation of questionnaire elements, enabling more adaptable and context-sensitive interviews. Additionally, the integration of multimodal recordings, together with automated transcriptions, reduces manual labor while supporting more expressive response formats (Barnwell, 2025). Notably, these systems allow perceived space to be manipulated, for instance through background modifications and avatar or voice transformations (Okano, Okada, Wakatsuki and Zempo, 2023; Han and Bailenson, 2024). This can increase accessibility (Creed, Al-Kalbani, Theil, Sarcar and Williams, 2024) and open up new design spaces for studying the effects of interview context on participant behavior (Notaro, Mariel and Hadjichristidis, 2024; Schmiedeberg and Schröder, 2024).

The second aspect concerns data capture and analysis. Audio and video recordings capture additional layers of information about the interview, e.g., shifts in participants' tone, emotional expressions, and levels of comfort (Barnwell, 2025). While such data could previously be collected through post-hoc assessments or local recordings, a system designed specifically to make use of these possibilities enables a more consistent and scalable acquisition of these signals. We therefore conceptualize interview data as a structured, networked, multimodal dataset in which relations between responses and contextual factors are explicitly represented.

Collectively, these developments motivate a shift from viewing interviews as simple mechanisms for response collection toward seeing them as systems for capturing and interpreting interconnected, context-rich interaction data. While XR systems provide a broad design space for materializing these shifts within a unified environment, their use is not yet widespread among the general population, especially compared to devices capable of video chats. As survey interview studies already struggle with participation, interview environments should not be limited to XR alone in order to ensure accessibility. Therefore, we propose a **unified** interview environment that supports the design and conduct of interviews in both XR and video-based modes.

To address these challenges, we present INTERVIEW, an extensible interview platform for XR and video-based environments. INTERVIEW builds directly on the VR-SUITE (Mehler, Bagci, Henlein, Abrami, Spiekermann, Schrottenbacher, Konca, Lücking, Engel, Quintino, Schreiber, Saukel and Zlatkin-Troitschanskaia, 2023) framework, which is based on hypertext principles and supports the generation and representation of multimodal hypertexts in VR. INTERVIEW's data model is explicitly designed to preserve relational structures across modalities. Analogously, questionnaire items, multimodal recordings, and contextual signals are modeled as nodes within INTERVIEW's data model, preserving their temporal and behavioral relations as typed links (Hargood, Hunt, Weal and Millard, 2016). INTERVIEW thereby unifies real-time multimodal communication, relational data structuring, and systematic data collection in a coherent environment, while enabling interview designs that leverage rich, interconnected multimodal data about participants and their interaction contexts.

The paper is structured as follows: We motivate a set of system requirements for a virtual-reality interview system in Section 2. Section 3 then provides an overview of the (survey) interview landscape and ongoing research on XR interview systems in relation to these requirements. We then present a set of use cases in Section 4 to illustrate the capabilities of a system such as INTERVIEW. Section 5 provides a detailed overview of our system architecture. We then evaluate the XR environment in Section 6 and present the results in Section 7. Finally, we present our conclusions and future work in Section 8.

2. System Requirements

Based on the introduction, we identify requirements for a flexible and extensible interview environment. A key concern for these also includes the need for accessible designs: the system should accommodate diverse user abilities, communication styles, and technical constraints. To this end, we extend and specify the set of criteria from (Mehler et al., 2023) as follows:

- (A) **Supports Questionnaires:** The system should integrate a software-based survey tool that allows interviewers to read questions and enter responses while presenting interviewees with the answer categories visually, thereby shaping the interview environment through its interface design.
- (B) **Avatar Customization:** The system should allow participants to customize their avatar-based appearance to tailor the experience to their preferences. This should include the ability to modify participants' voices. In addition to increasing perceived embodiment, this enables a form of user anonymization.
- (C) **Environment Customization:** The environment should be interchangeable and customizable for different interview needs.
- (D) **Multi-Platform Participation:** Participation in the interview should be possible across different device types, such as desktop computers, mobile devices, and VR headsets, including concurrent use.
- (E) **Platform Specific Multimodality:** The system should support platform-appropriate interaction modalities based on participants' requirements (e.g., gaze tracking for selecting entries). Moreover, the system should represent each participant to others as richly as possible using the available modalities.
- (F) **Data Collection & Linkage:** All data produced by the system should be systematically collected and stored in an accessible relational format that preserves the networked structure of the interview.
- (G) **LLM Assisted Interviews:** The system should support the use of Large Language Models (LLMs) to aid the interview process, whether for accessibility purposes (e.g., reducing language barriers) or for fully automated interviews conducted by embodied agents.

The next section reviews related work in light of our requirements, highlighting how current systems only partially satisfy them.

3. Related Work

Numerous studies have examined how user behavior in VR systems influences interaction and experience. A well-established phenomenon in avatar representation is the Proteus Effect (Yee and Bailenson, 2007), which describes how individuals change their behavior to conform to the characteristics of their avatar ((B)). A meta-analysis further confirms that this effect is a reliable phenomenon (Ratan, Beyea, Li and Graciano, 2020). Furthermore, avatar customization has been shown to affect self-perception and performance outcomes, with evidence that task performance can be influenced by self-representation (Ratan, Klein, Ucha and Cherchiglia, 2022). In immersive VR settings, these effects extend to emotional experience and embodiment. Radiah, Roth, Alt and Abdelrahman (2023) provide an overview of how avatar personalization affects users' perceived emotions, while Bujić, Macey, Kerous, Buruk and Hamari (2025) provide evidence that customization can increase users' agency. These effects are also relevant with regard to ethnic representation and matching, which can shape users' sense of embodiment (Do, Benjamin, Protko and McMahan, 2024). This further emphasizes the role of identity in social VR settings, where it is important to foster safe and comfortable environments (Wiik, Kähkönen and Smed, 2025; Quent and Vogl, 2025). Existing work primarily investigates these effects in isolation, without considering the potential for combined analysis.

Various XR and VR environments have been established with these effects in mind, focusing on immersive interview settings. Research has shown that VR-based interviews can approximate key aspects of face-to-face interactions, enhancing the perceived authenticity of the experience (Rzeszewski et al., 2024) (A). Beyond qualitative research contexts, VR has also been widely used in simulation-based interview environments. Luo, Wang, Lee, Xing, Jin, Dong, Hu, Chen, Yan and Hui (2024) explore the influence of realism, interview behavior, and question types on participants' anxiety and performance, while Kim, Sandhaus and Fussell (2023) examine gender-swapped

avatars in interview situations as a means of reducing anxiety and improving performance (A), (B), (C). Both studies highlight the potential of XR systems for controlled experimental studies. In addition, recent work has focused on behavioral and physiological measures in VR interviews, using eye-tracking and bio-signal analysis to better understand participant responses and social dynamics (Ni, Fang, Wang and Hui, 2025) (E). This further highlights the need for a unified and flexible process for data representation and collection. Furthermore, with regard to realism, Ashrafi, Vona, Ringsdorf, Hertel, Toni, Kailer, Bartels, Kojic and Voigt-Antons (2024) explore the use of photorealistic AI-driven avatars to support interviews across different settings (G). While these systems demonstrate the feasibility and benefits of immersive interview environments, they primarily focus on interaction design, user experience, and behavioral outcomes rather than on the structured capture and representation of interview data.

A rapidly growing research area concerns AI-assisted interview systems, ranging from text-based chatbots (Jacobsen, Cox, Griggio and van Berkel, 2025; Kim, Lee and Gweon, 2019) to fully embodied conversational agents (Ashrafi et al., 2024; Budig, Nißen and Kowatsch, 2025) (G). Kaiyrbekov, Dobbins and Mooney (2025) employ an LLM-powered conversational agent to conduct surveys and transcribe interactions, while Jacobsen et al. (2025) use LLM-based chatbots to support qualitative data collection through interview probes embedded in a web survey. Moreover, Wuttke, Aßenmacher, Klamm, Lang, Würschinger and Kreuter (2025) and Cuevas, Scurrrell, Brown, Entenmann and Daepf (2025) highlight the viability of generating high-quality data through AI conversational models at scale. However, these approaches remain largely constrained to textual or conversational representations and do not fully leverage the multimodal and contextual richness of interview environments (D).

Interviews are inherently multimodal, as shown by Dong, Santiago-Anaya and Jeon (2023) and Dubosc, Gorisse, Lefrou, Richir and Christmann (2025), who demonstrate that the level of facial expression stylization strongly affects user perception. Brambilla, Masi, Mattavelli and Biella (2021) specifically show the importance of alignment between an avatar's voice and facial appearance for perceived trustworthiness, which also aligns with findings by Kim et al. (2023). These findings highlight the role of multimodality in interviews, yet current systems lack the systematic collection and structuring of such data within a unified environment. This gap motivates the need for an interview system that not only supports immersive and multimodal interaction, but also enables the systematic collection and relational structuring of interview data within a unified environment.

4. Use Cases

To illustrate the breadth of INTERVIEW's applicability, we present a non-exhaustive set of use cases enabled by its unified multimodal design.

4.1. Simulation-based evaluations

As highlighted in Section 3, one of INTERVIEW's primary use cases is simulation-based evaluation. The ability to customize the virtual environment and avatars allows the simulation of different scenarios, such as job interviews or exams. This enables participants to train while allowing researchers to collect multimodal interaction data in realistic settings (Harris, Bird, Smart, Wilson and Vine, 2020). Depending on the operators' needs, this data can provide valuable insights into participants' behavioral patterns, e.g., the main gaze focal point during an interview or users' anxiety levels (Luo et al., 2024). These insights arise from linking behavioral signals to the simulation context, enabling richer post-hoc analysis.

4.2. Virtual Turing tests

Another use case of INTERVIEW is the implementation of virtual Turing tests (cf. Jones and Bergen, 2025). Within an interview environment, embodied AI agents can be introduced as interview partners for either the interviewer or the interviewee, without participants necessarily being aware of their presence. These interviews provide insights into both human-AI perception (Lim, Schmälzle and Bente, 2025; Rubo and Neumann, 2025; Vallefucio, Varni and Zancanaro, 2025) and conversational naturalness. While not evaluated in the present study, INTERVIEW's architecture is designed to support future implementations of embodied AI interview partners thus enabling the systematic introduction of such agents and providing insights beyond self-reported measures through the collection of networked behavioral signals.

4.3. Virtual discussion groups and consensus-building

Virtual discussion groups represent a third use case for INTERVIEW. Participants can be presented with shared stimuli or discussion topics within a customizable virtual environment, enabling structured group interactions

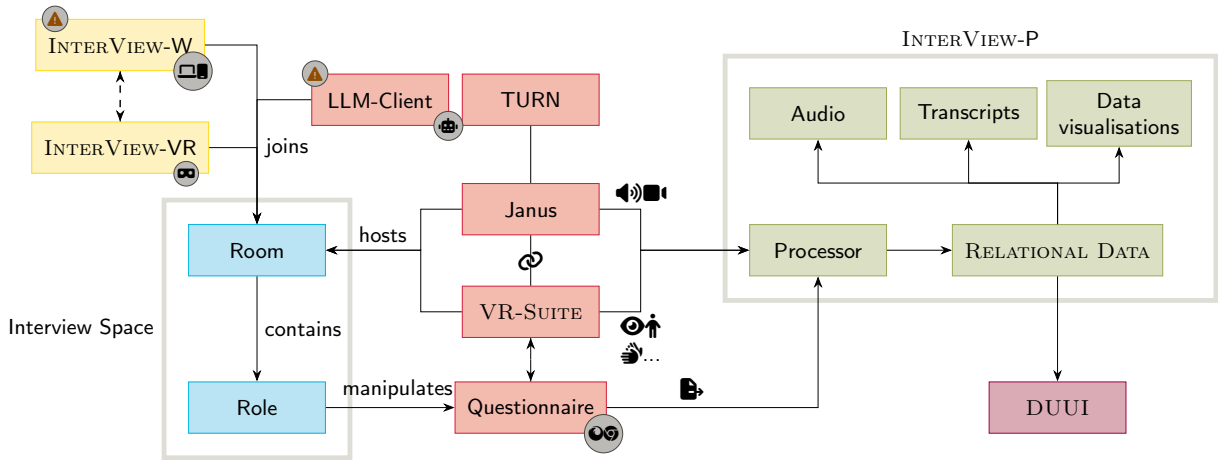


Figure 1: Overview of INTERVIEW's architecture: The various client types joining a room instantiated via VR-SUITE and Janus are shown in yellow in the top left. The clients are then assigned roles and can communicate with each other. Furthermore, they can interact with and modify the questionnaire associated with the environment. Data captured by VR-SUITE and Janus, together with questionnaire data, can be used by INTERVIEW-P to derive additional insights, with optional advanced processing through DUUI. Nodes denoted by orange exclamation marks have not been the focus of this study.

in a controlled setting. Traditional focus group methods primarily capture individual responses, with relational dynamics between participants remaining largely unobserved (de Souza, Gillett, Salifu and Walshe, 2024; Cyr, 2016). INTERVIEW addresses this gap by systematically capturing multimodal signals underlying these dynamics, including gaze patterns (Wang, Han, Queiroz, DeVaux and Bailenson, 2025) and turn-taking behavior (O'Connell, Kowal and Kaltenbacher, 1990; Sacks, Schegloff and Jefferson, 1974), two central mechanisms of group coordination (Degutyte and Astell, 2021; Mills and Boschker, 2022). INTERVIEW's networked data structure preserves these temporal and contextual links between the signals, allowing researchers to study the coordination process. This makes INTERVIEW particularly well suited for research on deliberation and conflict resolution or for any setting in which the process of arriving at a conclusion is as important as the outcome itself.

5. Architecture

INTERVIEW's architecture is divided into three main components: the backend (INTERVIEW-B), the frontend, in this case the VR-frontend (INTERVIEW-VR) and web-frontend (INTERVIEW-W) (D), as well as the interview processor (INTERVIEW-P). An overview can be seen in Figure 1.

The backend is based on three services: a TURN server (Mahy, Matthews and Rosenberg, 2010), implemented using coturn (Systems, 2026), which supports real-time audio and video transmission between clients; Janus (Amirante, Castaldi, Miniero and Romano, 2014; Amirante, Castaldi, Miniero and Saviano, 2017), a WebRTC server responsible for hosting the interview rooms and handling the initial connection negotiation; and VR-SUITE, which records multimodal data and uses Ubiq (Friston, Congdon, Swapp, Izzouzi, Brandstätter, Archer, Olkkonen, Thiel and Steed, 2021) as a networking library in Unity. Any survey tool or questionnaire can be embedded within the interview, provided it is accessible via a web browser (A). The VR-SUITE server forms the core of the backend and performs three main functions: it hosts rooms via Ubiq that participants can join, synchronizes the environment and avatars across participants and records all data produced in the environment (F). The VR-SUITE server is linked to the Janus service, ensuring that each room in VR-SUITE corresponds to a room in Janus. Each user joining a VR-SUITE-hosted room is assigned to the corresponding room in Janus. Janus is responsible for negotiating RTP/RTCP media streams between clients for audio and, optionally, video. Importantly, Janus provides a platform-independent video chat service that is easily extensible via plugins to support diverse requirements. While the use of a TURN service such as coturn is optional, it can be critical for establishing connections through network address translators (NATs) or firewalls commonly deployed by universities and other large institutions, where direct connections would otherwise not be possible (Mahy et al., 2010). The Janus service also handles the recording of media streams, which can be toggled on

or off on demand. Ubiq is responsible for synchronizing user states within the VR environment, including hand, face, body, and eye-tracking data, as well as the state of the questionnaire and other entities. VR-SUITE records these data in its database. While not yet fully realized, the backend also supports an LLM client that can be incorporated into INTERVIEW to assist with or fully conduct an interview (G).

INTERVIEW-VR is based on VR-SUITE, a multimodal XR framework that supports collaborative experiences with customizable avatars and environments (B),(C). It was extended with an OS-agnostic web browser serving as the main interface to the questionnaire (D). In addition to allowing one user to control the questionnaire, the system supports reactive events based on browser content, such as dynamically displaying answer options exclusively to the interviewee. INTERVIEW-VR also leverages Janus for media stream negotiation, enabling audio and video playback of other participants, e.g., via INTERVIEW-W. Participants using INTERVIEW-VR are represented by articulated avatars conveying facial expressions, gaze, and hand and body movements (E).

INTERVIEW-W is a static HTML site served via nginx that allows participants to join an interview room using a token and then requests access to the microphone and webcam. The JavaScript Janus protocol implementation is then used to negotiate connections between clients in the room, enabling the exchange of audio and video streams. The client also connects to VR-SUITE to synchronize elements across clients. Participants using INTERVIEW-W are represented either by a video feed or an emulated avatar in the environment (E). Both representations enable the collection of participants' gaze and facial data, although video-based capture is more limited (F).

INTERVIEW-P is a small standalone set of processes handling basic tasks such as collecting the data from the backend services for each interview that was conducted, alongside simple Automatic Speaker Recognition (ASR) tasks, through whisper (Radford, Kim, Xu, Brockman, Mcleavey and Sutskever, 2023) or other whisper-based models. Paired with INTERVIEW-VR, it also allows for the detection and visualization of gaze targets within the interview environment, as well as additional video-based recordings from within the virtual interview environment, e.g. for video-based emotion detection. Furthermore, questionnaire data from services such as LimeSurvey can be imported and synchronized (F). This allows relevant connections to be made between the questions and the recorded multimodal interview data. The resulting RELATIONAL DATA can further be processed by DUUI Leonhardt, Abrami, Baumartz and Mehler (2023), a multimodal corpora processing pipeline capable of distributed computation of various metrics through hundreds of extendable tools. The relational structure of the data collected by INTERVIEW-P instantiates the principles outlined earlier: instead of questionnaire items, multimodal data and contextual metadata being stored as isolated entries, they are stored as interconnected nodes preserving the networked relationships that otherwise would be discarded.

6. Evaluation

To evaluate INTERVIEW, we conducted interviews in VR to analyze user behavior and derive insights into responses and emotional reactions to key parameters, including the participant's own avatar, the perception of the interviewer's avatar (including the congruence of voice and appearance), openness when answering sensitive questions, and interaction-related factors. Participants were able to select one of four avatars to be represented by. To derive this set of avatars an online experimental study was conducted to identify suitable avatar representations for large-scale survey applications. To this end, we conducted an initial Computer Assisted Web Interview (CAWI) survey in which participants were exposed to 20 unique avatars. The rather cartoonish avatars were designed to systematically vary key characteristics, including gender, age, and cultural background. To ensure a credible representation of the interviewer role, all avatars depicted adults with a well-groomed appearance. The survey was designed to investigate how different respondent groups selected avatars to represent interviewers. Participants evaluated each avatar on several dimensions, most notably perceived trustworthiness and their anticipated level of comfort during an interview. In addition, respondents first identified up to three avatars they considered most similar to themselves and subsequently ranked their three preferred interviewer avatars. A total of $N = 98$ participants completed the survey. From these 20 avatars, we selected two of different genders that ranked highly (Category A) and two that ranked poorly (Category B) in our study, in regard to user preference. The selected avatars can be seen in Figure 2. Based on participants' avatar selections, we assigned a different avatar from the same set of four options to ensure a balanced distribution across conditions. Importantly, interviewees neither saw nor knew the interviewers beforehand, as the interviews were conducted in two separate institutions. The survey questions ranged from typical inquiries about participants' education, income, and relationship status to their offline and online habits, as well as items on personality traits (Back, Küfner, Dufner, Gerlach, Rauthmann and Denissen, 2013; Henning, 1983; Bosnjak, Metzger and Gräf, 2010).



Figure 2: The four avatars used during the interview. The first two avatars (A_1, A_2) are part of category B, the less desirable avatars, while the second two (A_3, A_4) being part of category A.

While some questions were benign, others were designed to elicit discomfort among participants. The interview lasted about 20 minutes and included two open-ended questions prompting participants to share details about their personal lives. The interviewee was seated at a desk facing the interviewer in a medium-sized room. A narrow display on the desk showed the categorical answer options for the current question, as well as a small video feed on the far right that served as a mirror for the interviewee. The interviewer had a large screen displaying the LimeSurvey questionnaire, with which they could interact via touch, hand gestures, controllers or gaze. During the interviews, the system recorded multimodal data, including audio and interaction data such as gaze and avatar movements. Behavioral changes were then analyzed across verbal responses (e.g., response length and hesitations), non-verbal cues, and patterns of self-disclosure. Furthermore, we administered a post-interview questionnaire to participants and the interviewer. Specifically, the interviewee questionnaire included items adapted from UMUX (Finstad, 2010) to assess system usability, as well as items on comfort within the environment and reasons for avatar choices. The interviewer questionnaire included items assessing technical difficulties encountered during the interview. It also assessed the interviewer's perception of the interview flow and the integration of answer options within the environment. The goal of this experiment was to assess participants' acceptance of the interview environment and their (dis)comfort with the questions and avatar choices. We pose the following research questions:

1. RQ_1 : Does the embedded questionnaire interface change response behavior?
2. RQ_2 : Do multimodal signals index question sensitivity?
3. RQ_3 : Do multimodal signals converge with or diverge from self-report?

7. Results & Discussions

In total, we conducted $N=27$ interviews, with male interviewers in 19 cases and female interviewers in 8 cases. Among the interviewees, the gender was male in 19 cases and female in 7, with one case not disclosed. The interviewees selected avatars A_1, A_3 , and A_4 in 3, 11, and 13 cases, respectively, while the interviewer was represented by A_1 - A_4 in 8, 8, 5, and 6 cases, respectively. Importantly, avatar A_2 was never chosen by participants, potentially due to its culturally specific appearance, which may not have matched participants' self-perceptions or identification preferences. Further supported by the pre-study which sampled from a similar background of participants. Interviewees selected avatars matching their self-reported gender in 21 out of 26 cases, while interviewer avatars matched the interviewees' gender in 15 out of 27 cases. Unless otherwise indicated, all scales are 5-point Likert scales, where 1 indicates low agreement and 5 indicates high agreement. Reported p -values for group comparisons use the Kruskal-Wallis test (Kruskal and Wallis, 1952) and within-participant paired comparisons use the Wilcoxon signed-rank test (Wilcoxon, 1945). Specifically, participants reported very low levels of discomfort ($\bar{x} = 1.48 \pm 0.79$) and dizziness ($\bar{x} = 1.39 \pm 0.84$) during the interview. Participants also reported moderate enjoyment ($\bar{x} = 3.39 \pm 0.66$) and a high willingness to participate in future studies ($\bar{x} = 4.65 \pm 0.71$), with most describing the overall experience as very positive ($\bar{x} = 4.30 \pm 0.76$). Most

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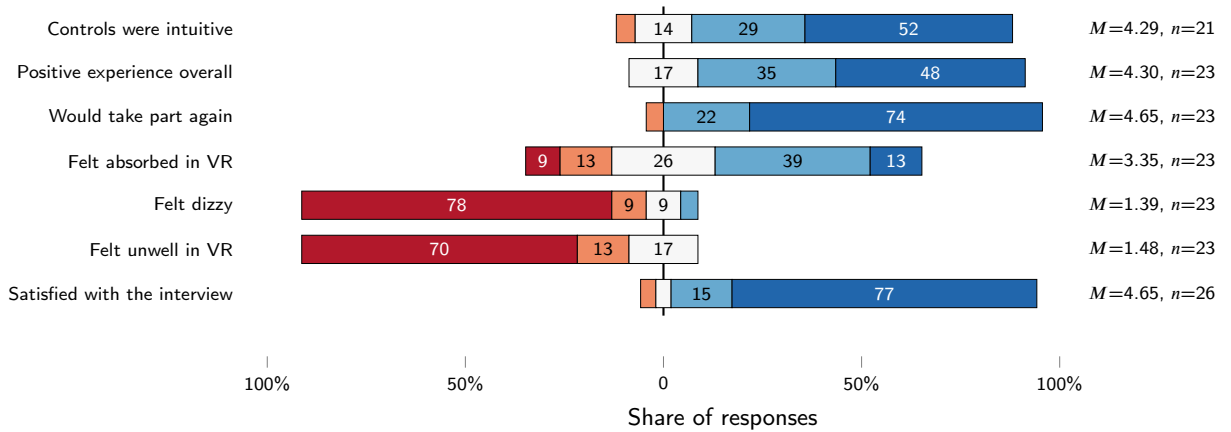


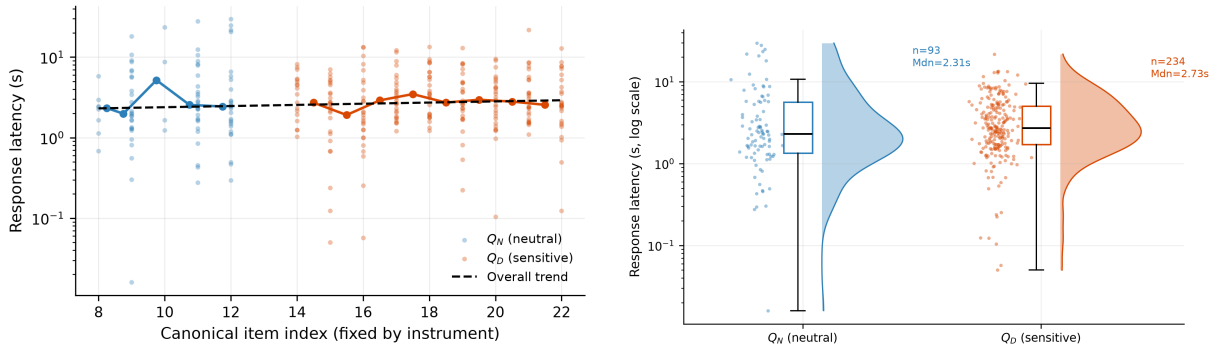
Figure 3: Overview of the self-reported user experiences with the application environment. The questions were 5-point Likert scales ranging from strongly disagree in red to strongly agree in blue. Overall, their experience with the system was evaluated positively.

participants reported at least moderate immersion in the environment ($\bar{x} = 3.47 \pm 1.15$). A breakdown of these numbers can be seen in Figure 3. These measures are independent of the interviewer's chosen avatar and avatar-participant matching, indicating a stable and robust user experience across conditions. These results suggest that INTERVIEW provides a feasible and usable VR-based interview setting.

To obtain further insights into the respondents' behavior and acceptance of the environment we specifically introduced sensitive questions into our questionnaire. Based on these questions we conducted an exploratory analysis of selected interview excerpts to examine behavioral responses to sensitive questions. Specifically, we compared two groups of questionnaire items regarding personality traits, one neutral (Q_N) and the other discomfort-inducing (Q_D), which were both Likert-scale items served as the baseline to obtain an insight into the participants' perception of the interview.

7.1. RQ_1 : Does the embedded questionnaire interface change response behaviour?

The interviewer's screen and the interviewee's answer-option panel are the two points at which INTERVIEW's questionnaire logic is directly exposed inside the immersive environment. Interviewers unanimously reported that the portrayal of the answer options aided in the flow of the interview, already providing a first insight as to the usage and usability of such an element. Gaze analysis further identified a clear hotspot on the panel displaying the answer options, especially when the interviewees needed to decide on a response. First we looked at the individual verbal responses by classifying how each spoken answer referenced an n-point scale (if applicable): as a bare index, as the full textual label, as both or as neither. Overall, 36.3% of responses used the index alone and 32.6% used the full label, with 4.4% combining both and 26.8% using neither. The last result can be partially attributed to the automated transcription that was used not being perfect but in most cases, respondents modified the response format, for example by using abbreviated versions of the provided labels. A comparison of index-form usage between sensitive and neutral items did not reach significance (Wilcoxon signed-rank on per-participant index-share, $W=70.0$, $p=0.758$, $d_z=0.10$, $n=26$), indicating this behavior to be a property of the interface as a whole rather than question sensitivity. Additionally, we compared index and label usage across Likert scales of different lengths. While we observed differences in index usage between scale types, these differences did not follow a monotonic pattern with increasing scale length. A Friedman test indicated differences in index-answer share across the scales (χ^2 -statistic = 12.09, $p=0.0024$, $n=26$). However, these differences are unlikely to be attributable to scale length itself - the 4- and 6-point items are otherwise-unrelated agreement scales (epistemic curiosity; narcissism/Machiavellianism), whereas the 5-point items are frequency scales (locus of control; activity frequency, e.g. *daily/never*). The distinct answer-option wording of the frequency scales, rather than the number of options, is the more plausible explanation for their lower index usage. Notably, these indices were only visible within the question panel and not read aloud; participants could only have produced them by reading the panel and incorporating them into their response. Importantly, this highlights the potential challenges of automating answer deriving processes based on the spoken word of the interviewee, since 26.8% of the answers would be missed



(a) Response latency by item position within the interview. No monotonic drift with position is apparent for either question type. (b) Turn-taking latency by question type. Neutral questions tend to be answered a bit slower than sensitive ones.

if relying solely on resolving answer indexes and labels from the spoken word of the interviewees. This demonstrates the impact that an answer-option interface can have on the interview itself. It clearly changes the participants' response behavior, as contextualized by the multimodal data collected by INTERVIEW, whilst simultaneously aiding the flow of the interview.

7.2. RQ_2 : Do multimodal signals index question sensitivity?

Response Latency: We observed an additional 0.42 seconds of latency on average (2.73s vs 2.31s for neutral items) for the responses from the interviewees when confronted with questions from the Q_D group, although this difference is not statistically reliable (4(b)).

Critically, the two batteries were not interleaved: all Q_N items precede all Q_D items, so any effect of item position is entangled with question type. To separate the two, we fit a cluster-robust log-latency model with question type and relative interview position as predictors. Neither term is reliable (sensitivity factor 0.89, 95% CI [0.58, 1.38], $p=0.613$; position factor 1.48, 95% CI [0.88, 2.48], $p=0.143$), and 4(a) shows no clear monotonic drift across the interview. Notably, the model's sensitivity estimate points in the opposite direction from the raw comparison above, suggesting that the raw latency difference is better explained by item position - and thus potentially fatigue or habituation - than by question sensitivity itself. We therefore do not claim a latency effect of either sensitivity or position: once position is accounted for, question type alone does not separate the two conditions. A log-latency model with question type and relative interview position as predictors confirms this: neither term is statistically reliable. This indicates that the response latency trends upwards throughout the duration of the interview with the question type not adding a statistically separable effect. **Gaze:** We defined areas of interest (AOIs) within in the Unity interview scene which allowed us to analyze the interviewees gaze behavior by replaying the previously captured data within the environment, gathering the points the participant looked at within the 3D geometry. Dwell share per condition is shown in Figure 4. No AOI shift survives Holm correction for multiple comparisons. The largest raw shift is an increase in dwell on the window (+0.026, $p = 0.162$ uncorrected); the self-view mirror (-0.003 , $p=0.433$), the answer options panel (-0.009 , $p=0.627$), and the interviewer's body (-0.005 , $p=0.492$) are essentially unchanged, and dwell time on the interviewer's face itself barely moves (+0.001, $p=1.0$). Beyond dwell location, we examined fixation dynamics using an I-VT threshold (Salvucci and Goldberg, 2000) of $20^\circ/s$. The fraction of exposure time spent fixating shows a reliable decrease under Q_D (0.63 vs. 0.58 s/s, Wilcoxon $p=0.008$, $d_z=-0.64$, $n=26$) that survives Holm correction across the fixation-metric family ($p_{Holm}=0.040$). Within the same Holm-corrected family, fixation rate (1.79 vs. 1.82 fixations/s, $p=0.940$), mean fixation duration (0.34s vs. 0.29s, $p_{Holm}=0.117$), and gaze dispersion (13.1° vs. 14.6° , $p=1.0$) did not reach significance; fixation time-share is the only dynamic that reliably separates the two conditions (Figure 6). Beyond how often and how long participants fixated, we also asked how many distinct regions their gaze visited - its spatial entropy over a coarse grid of look directions. Entropy is higher under Q_D than Q_N (2.98 vs. 3.17, Wilcoxon $p=0.018$, $d_z=0.57$, $n=20$), consistent with scanning across more of the scene under sensitive questions rather than the wider radial staring that dispersion alone would capture. This is also visible in Figure 5. This contrast is significant from a bin width of 16 cells upwards ($p=0.10 - 0.040$). **Speech:** We further analyzed the response length for sensitive items and identified a longer median duration in comparison (1.51s vs 2.02s, Wilcoxon $p=0.025$). Response length in

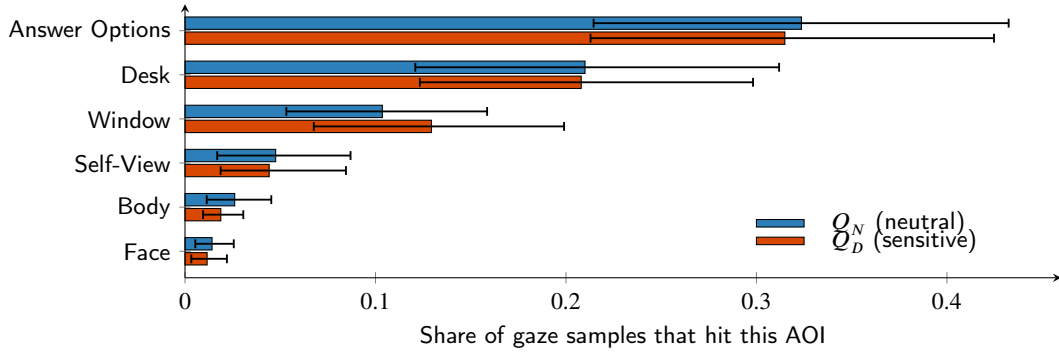


Figure 4: Mean dwell share per area of interest (AOI), by question type, with 95% bootstrap confidence intervals. Answer options and the interviewer’s desk dominate gaze allocation for both conditions; no AOI shift survives correction for multiple comparisons.

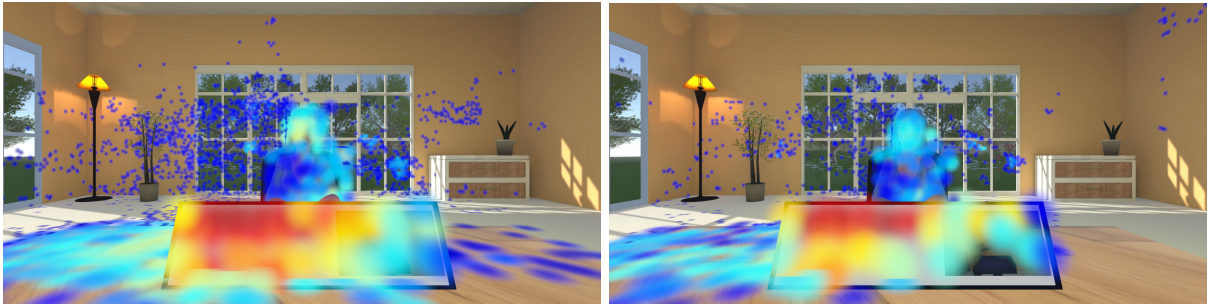


Figure 5: An example of a scene from the interview space, with an overlay showing the gaze hotspots of the participants between the two question groups. The image on the left shows the gaze hotspots for group G_D , which included questions designed to cause some discomfort, compared to group G_N , which did not include such questions. The gaze hotspots for group G_N are visualized in the image on the right. Red regions denote areas on which participants focused their gaze more, while blue denotes areas on which participants focused their gaze less. Since G_D included more survey items and took longer to answer, more gaze points are available for this group. One can, however, notice that the overall area containing actual gaze hotspots is slightly larger for G_D .

words, articulation rate showed no reliable difference between the conditions. In conclusion, gaze and speech show directional shifts consistent with question sensitivity, while the latency effect is confounded with item position and does not survive adjustment. We present this as what a fine-grained, timestamp-synchronized multimodal pipeline makes visible and falsifiable, not as a confirmed discovery.

7.3. RQ_3 : Do multimodal signals converge with or diverge from self-report?

Only 10 of the 27 respondents (37.0%) reported having felt discomfort with certain questions at all when asked directly in the secondary survey, but the behavioral shifts analyzed under RQ_2 are broadly present regardless of this self-assessment. Participants who claimed discomfort show a significantly larger increase in response duration from Q_N to Q_D . (Figure 7; Mann-Whitney $U=158.0$, $p_{Holm}=0.001$, $d=1.76$, $n_{yes}=10$, $n_{no}=17$). They also show a larger increase in dwell on the interviewer’s face ($U=128.0$, $p_{Holm}=0.037$, $d=0.73$, $n_{yes}=10$, $n_{no}=16$), while latency and mirror dwell time do not distinguish between the two groups. Overall, self-reported discomfort and observed behavior showed only partial convergence. This convergence was evident for response duration and face dwell time, but not for response latency. Participants who explicitly reported discomfort displayed the strongest behavioral changes, particularly longer responses and increased dwell time on the mirror, suggesting that these measures are sensitive to question-induced discomfort. At the same time, similar behavioral shifts were observed among participants who did not report discomfort when asked directly. If these behavioral changes are interpreted as indicators of discomfort, this finding suggests that such experiences may not always be captured through self-report alone. One possible explanation is that respondents may be reluctant to acknowledge discomfort explicitly, for example due to social desirability concerns. Alternatively,

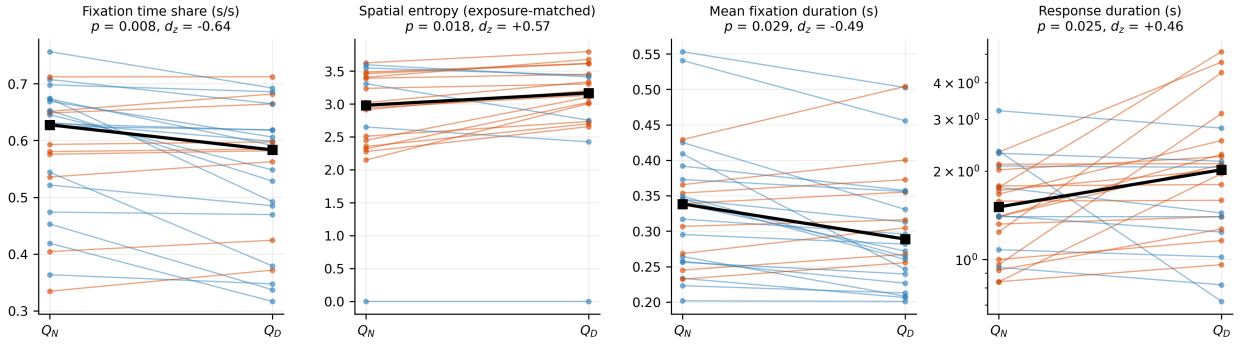


Figure 6: Per-participant paired fixation and speech metrics under Q_N and Q_D : mean fixation duration, fixation time share, and spatial entropy (exposure-matched) and response duration. Black squares mark condition means; thin lines connect individual participants.

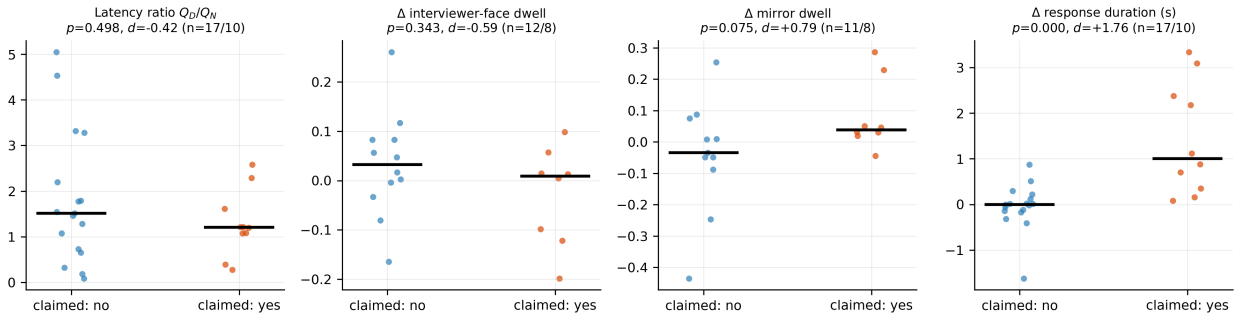


Figure 7: Measured behavioral shift ($Q_N \rightarrow Q_D$) split by whether participants claimed discomfort with certain questions in the secondary survey.

participants may not fully recognize or recall their momentary reactions when reflecting on them after the interview. Consequently, relying exclusively on self-reported discomfort may substantially underestimate the impact of sensitive interview questions.

7.4. Summary & Limitations

Crucially, these findings were identifiable because the system preserved the relational connections between question context, temporal position in the interview, and multimodal data signals. This directly reflects the networked data structure that motivated our hypertext-informed design.

Across our evaluation, we find that INTERVIEW supports the execution of VR-based interviews in a usable and stable manner, enabling the integration of questionnaire elements directly into the immersive environment. Furthermore, our results indicate that, beyond usability, multimodal signals can be systematically captured and aligned with interview context. This allows responses to be analyzed as part of a structured interaction process, not only as isolated answers. This finding supports our overarching goal of treating interviews as relational, multimodal data systems. This capability also raises an ethical consideration. INTERVIEW's behavioral signals were associated with discomfort that the majority of participants did not disclose when asked directly. Such behavioral inference can reveal information that participants did not intend to share, raising questions about consent and how these signals should be used, particularly if the underlying detection is improved in future work. Future applications of this capability should therefore treat transparent disclosure and careful data handling as requirements, not afterthoughts.

Our study has several limitations. First, the relatively small sample size ($N = 27$) limits the generalizability of the statistical findings. Second, the evaluation is exploratory and does not include a controlled comparison with traditional interview methods, such as video or text-based interviews. Third, although the system supports the capture of multimodal data, the present evaluation primarily focuses on the VR-based interview component. Additionally, the surveys were conducted in German so generalizability across languages is untested and since some of the evaluations

are based on transcripts derived from ASR systems so some data quality limitations exist. Finally, observed behavioral patterns should be interpreted as exploratory findings rather than causal effects.

8. Conclusion & Future Work

We proposed INTERVIEW, an extensible, unified XR interview environment that integrates real-time communication, immersive interaction design, and structured multimodal data capture within a single system. INTERVIEW is open source and publicly available on GitHub¹, supporting extensibility and reusability. While our evaluation provides valuable insights, it focuses on the VR-headset condition; future work will extend the evaluation to the web-based client and cross-platform interaction scenarios, particularly investigating seamless interaction between VR, desktop, and mobile clients. To this end a larger study is currently ongoing. In this context, expanding the web client's feature set to support avatars and dynamic backgrounds represents a key direction. Additionally, avatar representation should be explored further, including other types of avatars, especially more realistic ones, in combination with the capability for voice transformation. Such extensions would allow for additional analyses of how identity, presence, and social cues influence behavioral patterns in interviews. Support for an LLM client, as well as additional biosensors (e.g., EEG), will introduce further possibilities for interview designs and evaluations for INTERVIEW. Future research will also explore how the structured multimodal data captured by INTERVIEW can be systematically leveraged for interview analysis. Overall, our findings emphasize the value of reframing interviews as interactive hypertext systems that capture and integrate multiple dimensions of human behavior within a unified environment.

Data availability

Due to the sensitivity of the data collected and the agreement the respondents signed the data can not be shared.

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Ethics Statement: The study was approved by the ethics committee of the Leibniz Institute of Educational Trajectories. All participants provided informed written consent before participating. Participants were informed that gaze, facial expressions and audio would be recorded and stored in a secure database. Participants could withdraw at any time without consequence complying with the GDPR.

CRedit authorship contribution statement

Patrick Schrottenbacher: Software, Methodology, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft. **Alexander Mehler:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing. **Giuseppe Abrami:** Software, Resources, Writing – original draft. **Lydia Kleine:** Methodology, Investigation, Resources. **Doris Stingl:** Methodology, Investigation, Resources.

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¹<https://github.com/texttechnologylab/Va.Si.Li-Lab>

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

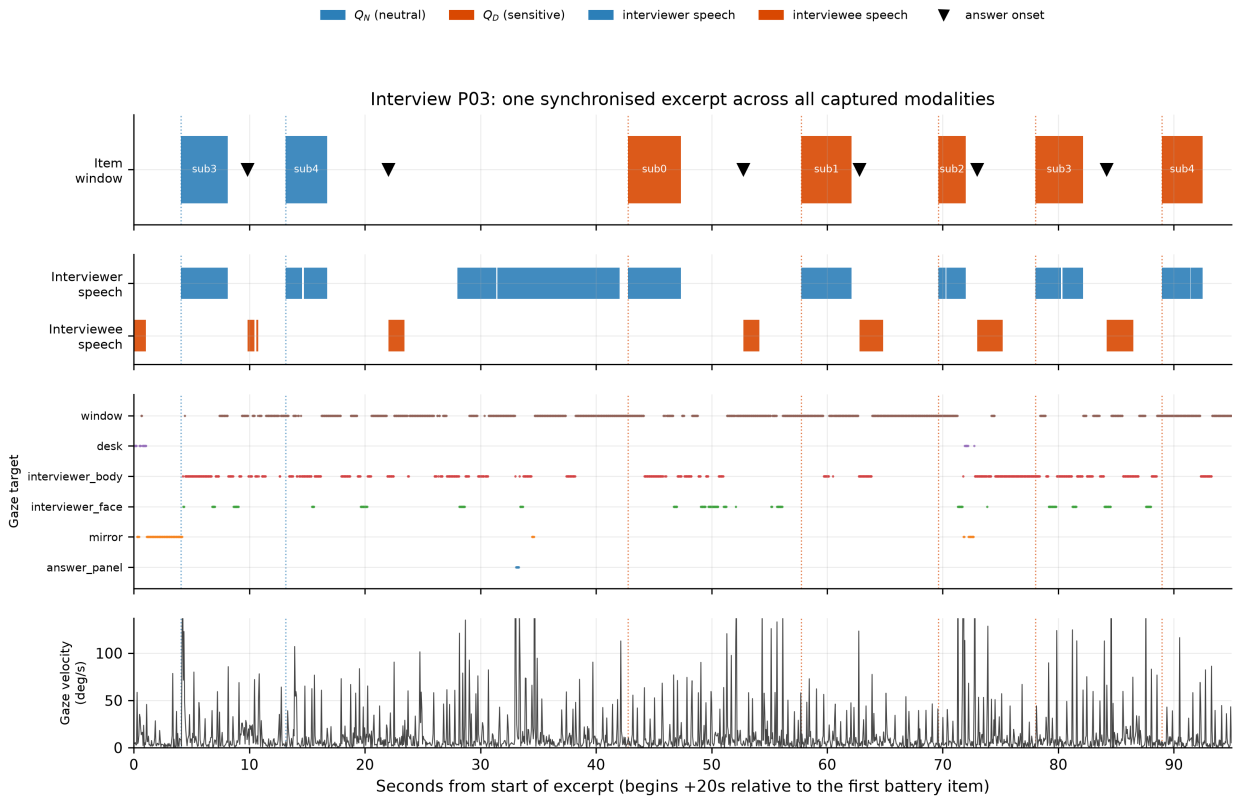


Figure 8: One synchronized excerpt from a single interview across all captured modalities.